

Chhattishgarh Swami Vivekanand Technical University-UTD

Department of Computer Science & Engineering

Cryptography & Network Security Notes, Unit-1

UNIT-1: Classical Encryption Techniques

1. Introduction to Cryptography

1.1 What is Cryptography?

The word **Cryptography** is derived from the Greek words:

- **Kryptos** – hidden
- **Graphein** – writing

Thus, cryptography literally means **hidden writing** or **secret writing**.

Definition: Cryptography is the science and practice of protecting information by transforming readable data (**plaintext**) into an unreadable form (**ciphertext**) using an encryption algorithm and a key, so that unauthorized users cannot understand the information.

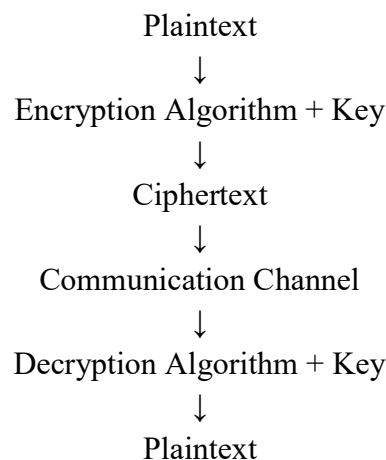
- **Plaintext:** Original readable message.
- **Ciphertext:** Encrypted form of the message.
- **Encryption:** Conversion of plaintext into ciphertext.
- **Decryption:** Conversion of ciphertext back into plaintext.
- **Key:** Secret value that controls the cryptographic operation.

1.2 Basic Example

Suppose Alice wants to send:

'MEET ME AT 10 AM'

The basic process is:



Only a receiver possessing the appropriate key can recover the original message.

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1.3 Need for Cryptography

Modern communication uses public networks such as the Internet. During transmission, information may be intercepted, modified, disclosed, or misused. Cryptography provides protection against these threats.

Common applications include:

- Online banking and UPI
- ATM and card transactions
- Email and messaging
- Digital signatures
- Cloud computing
- Healthcare systems
- Government digital identity systems
- Blockchain
- Artificial Intelligence applications

1.4 Objectives / Security Services

1. Confidentiality

Ensures that information is accessible only to authorized users.

Example: Banking credentials should not be readable by unauthorized persons.

2. Integrity

Ensures that information is not altered without authorization.

Example: Examination marks should not change while being transmitted to a university server.

3. Authentication

Verifies the identity of a communicating entity.

Example: Login through a password or biometric authentication.

4. Non-Repudiation

Provides evidence so that a sender cannot later deny an action or message.

Example: A digitally signed document provides evidence of the signer's involvement.

5. Availability

Ensures that authorized users can access information and services when required.

Example: Online banking should remain available to legitimate customers.

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1.5 Cryptography in Artificial Intelligence

AI systems process sensitive information such as medical records, financial data, biometric information, autonomous-vehicle data, and training datasets. Cryptography helps protect such information during storage, processing, and communication.

Important applications include:

- Healthcare AI and medical-image protection
- Secure AI model deployment
- Privacy-preserving machine learning
- Federated learning
- Protection of training datasets
- Secure cloud-based AI services
- Secure communication with autonomous systems

1.6 Real-Life Example: UPI Transaction

Consider a ₹50,000 UPI transaction:

Customer

↓

UPI Application

↓

Encryption / Secure Communication

↓

Internet

↓

Bank Server

↓

Decryption / Verification

↓

Transaction Processing

Without adequate security, attackers could attempt to obtain credentials, modify transaction information, or disclose personal data.

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1.7 History of Cryptography

Cryptography has evolved over thousands of years.

Year	Development
1900 BC	Egyptian hieroglyphic secret writings
50 BC	Caesar Cipher developed by Julius Caesar
Middle Ages	Monoalphabetic and polyalphabetic ciphers
World War II	Enigma Machine used by Germany
1977	DES (Data Encryption Standard) introduced
1976	Public Key Cryptography proposed by Diffie and Hellman
2001	AES (Advanced Encryption Standard) adopted
Present	AI-assisted security, quantum-resistant cryptography, blockchain security

1.8 Major Applications

Banking: ATM, UPI, NEFT, RTGS, cards, online banking

E-Commerce: Secure transactions and payment protection

Communication: Email and messaging

Cloud: Secure storage and communication

Healthcare: Electronic health records, medical images, telemedicine

Government: Digital identity, Aadhaar-related systems, tax and passport systems

AI: Secure datasets, privacy-preserving learning, secure deployment

IoT: Smart homes, wearables, and smart-grid communication

1.9 Challenges

- Secure key distribution
- Key management
- Increasing computational requirements
- Quantum-computing threats
- Insider attacks
- Human error
- Weak password practices

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1.10 Cryptographic Terminology

Term	Meaning
Plaintext	Original readable message
Ciphertext	Encrypted unreadable message
Encryption	Conversion of plaintext to ciphertext
Decryption	Conversion of ciphertext back to plaintext
Cipher	Mathematical algorithm used for encryption
Key	Secret value controlling encryption and decryption
Cryptography	Science of secure communication
Cryptanalysis	Study of breaking encryption
Cryptology	Combination of cryptography and cryptanalysis

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2. Symmetric Cipher Model

2.1 Introduction

A **symmetric cipher** uses the same secret key for encryption and decryption. It is also called:

- Secret-key cryptography
- Private-key cryptography
- Single-key encryption

The sender and receiver must possess the same secret key.

2.2 Real-Life Analogy

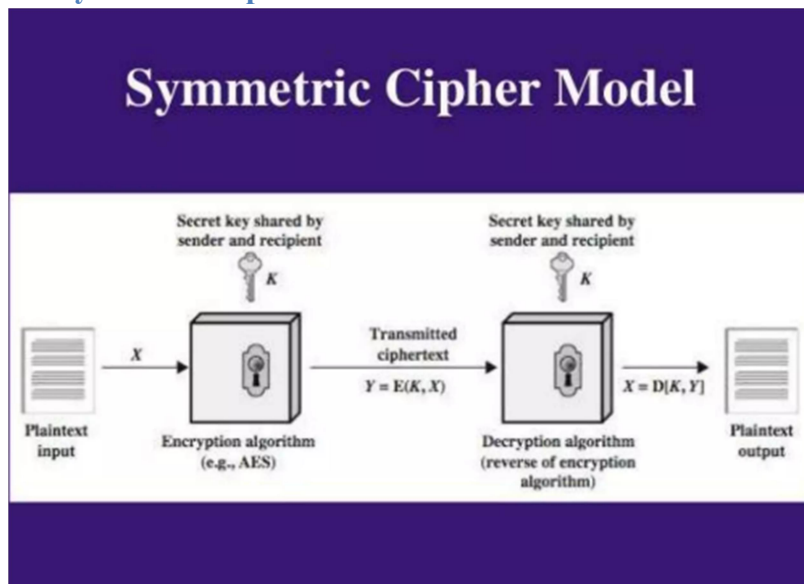
Consider a locker with one key:

Alice —[Lock using Key]—► Locker

Bob ◄—[Unlock using same Key]—

Both parties need the same key. Symmetric encryption follows the same basic idea

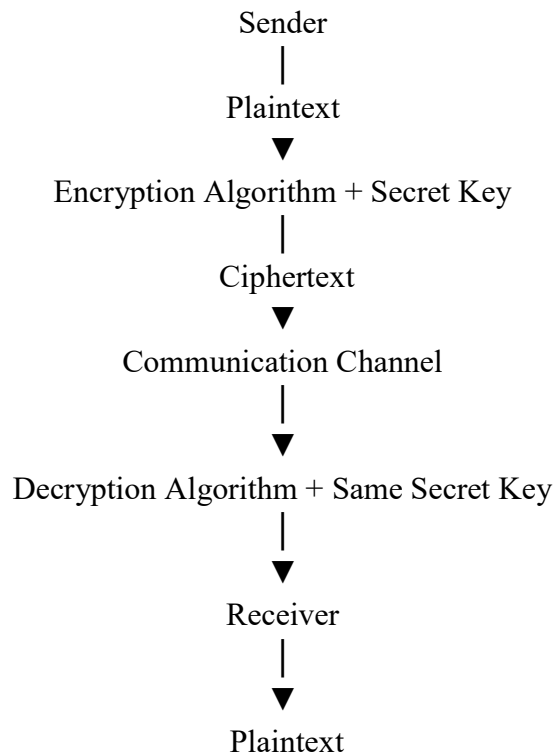
2.3 Symmetric Cipher Model



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Working Steps

1. The sender prepares plaintext.
2. The encryption algorithm uses the secret key.
3. Ciphertext is produced.
4. Ciphertext is transmitted.
5. The receiver applies the decryption algorithm.
6. The same secret key is used.
7. The original plaintext is recovered.

2.4 Components

1. Plaintext

Original readable information.

Examples: 'HELLO', 'PASSWORD', 'BANK ACCOUNT', or an AI dataset.

2. Encryption Algorithm

Transforms plaintext into ciphertext using the key.

Examples include AES, DES, Triple DES, Blowfish, IDEA, and RC4.

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3. Secret Key

A secret bit sequence known to the communicating parties.

Example:

'Key = 101011001101001'

The security of the system depends heavily on protecting this key.

4. Ciphertext

Encrypted output that should not reveal the original message to an unauthorized observer.

5. Decryption Algorithm

Uses the secret key to recover plaintext from ciphertext.

2.5 Mathematical Representation

Encryption:

$$C = E(K,P)$$

where:

- P = plaintext
- K = secret key
- E = encryption algorithm
- C = ciphertext

Decryption:

$$P = D(K,C)$$

where (D) is the decryption algorithm.

2.6 Example

Suppose:

Plaintext : HELLO

Key : SECRET

The encryption algorithm produces ciphertext. The receiver applies the corresponding decryption operation with the same secret key to recover:

'HELLO'

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The exact ciphertext depends on the algorithm and mode used; therefore, a ciphertext example should not be treated as universal for all symmetric algorithms.

2.7 Characteristics

- One shared secret key
- High speed
- Efficient for large amounts of data
- Low computational overhead compared with public-key operations
- Suitable for large files and high-throughput systems
- Requires secure key distribution and management

2.8 Advantages

- **High speed:** Suitable for high-volume data encryption.
- **Low computational cost:** Efficient for constrained devices.
- **Large-data efficiency:** Useful for video, databases, backups, and cloud storage.
- **Low memory overhead:** Generally efficient in resource-constrained environments.
- **Wide adoption:** Used in network security, storage protection, VPNs, and cloud systems.

2.9 Disadvantages

Key Distribution Problem

The sender and receiver must obtain the shared key securely.

Scalability

For a system with many users, managing separate shared keys becomes increasingly difficult.

Limited Support for Non-Repudiation

Symmetric encryption alone does not provide the same non-repudiation and digital-signature functionality associated with asymmetric cryptography.

Key Compromise

If the shared key is compromised, information protected with that key may be exposed.

2.10 Security Depends On

- Strength of the algorithm
- Key length
- Quality of randomness
- Secure key distribution
- Proper key management

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For an 'n'-bit key, the number of possible keys is:

Example

Key Size Possible Keys

40-bit 1 trillion

56-bit 72 quadrillion

128-bit 3.4×10^{38}

192-bit 6.27×10^{57}

256-bit 1.15×10^{77}

A larger key space generally increases resistance to brute-force search, although overall security also depends on the algorithm and implementation.

2.11 Popular Symmetric Algorithms

Algorithm	Key Size	Block Size	Status
DES	56 bits	64 bits	Obsolete
Triple DES	168 bits	64 bits	Legacy
AES-128	128 bits	128 bits	Recommended
AES-192	192 bits	128 bits	Highly Secure
AES-256	256 bits	128 bits	Very Highly Secure
Blowfish	32–448 bits	64 bits	Popular
IDEA	128 bits	64 bits	Secure

2.12 Applications

- Banking and payment systems
- Cloud storage
- Healthcare databases
- IoT devices
- Military and satellite communication
- AI datasets and model storage
- Secure communication and network protocols

2.13 AI Case Study: Secure Medical Diagnosis

A hospital stores CT or MRI images in a cloud system where AI models analyze them.

Risks:

- Unauthorized access
- Interception during transmission
- Disclosure of patient information

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Approach:

- Encrypt data using a strong symmetric algorithm such as AES.
- Restrict key access to authorized users and systems.
- Apply secure key management.

Result: Confidential medical information receives protection while remaining suitable for large-scale processing.

2.14 Symmetric vs Asymmetric Encryption

Feature	Symmetric Encryption	Asymmetric Encryption
Keys Used	One shared secret key	Two keys (public and private)
Speed	Very fast	Slower
Suitable For	Large volumes of data	Key exchange, digital signatures
Computational Cost	Low	Higher
Key Distribution	Difficult	Simpler through public keys
Examples	AES, DES, Blowfish	RSA, ECC, ElGamal

In practical systems, asymmetric cryptography is often used for authentication or key establishment, while symmetric cryptography protects bulk data.

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3. Steganography

3.1 Introduction

Cryptography protects the content of a message by transforming it into ciphertext.

Steganography instead attempts to conceal the **existence** of the message.

A secret message is embedded inside an apparently ordinary **cover medium**, such as an image, audio file, video, or text document.

3.2 Meaning and Definition

The word steganography comes from:

- **Steganos** – covered or hidden
- **Graphia** – writing

Definition: Steganography is the technique of hiding secret information inside another digital object so that the existence of the hidden information is not obvious to unauthorized observers.

3.3 Historical Background

Examples include:

- Secret writing in ancient Greece
- Invisible ink
- Microdots during wartime
- Modern hiding of information in digital images, audio, video, text, and network communication

3.4 Basic Terminology

Term	Description
Secret Message	The confidential information to be hidden.
Cover Medium	The carrier file (image, audio, video, or text).
Stego Object	The cover medium after embedding the secret message.
Embedding	The process of hiding the message.
Extraction	The process of recovering the hidden message.
Stego Key	An optional secret key used to control embedding and extraction.

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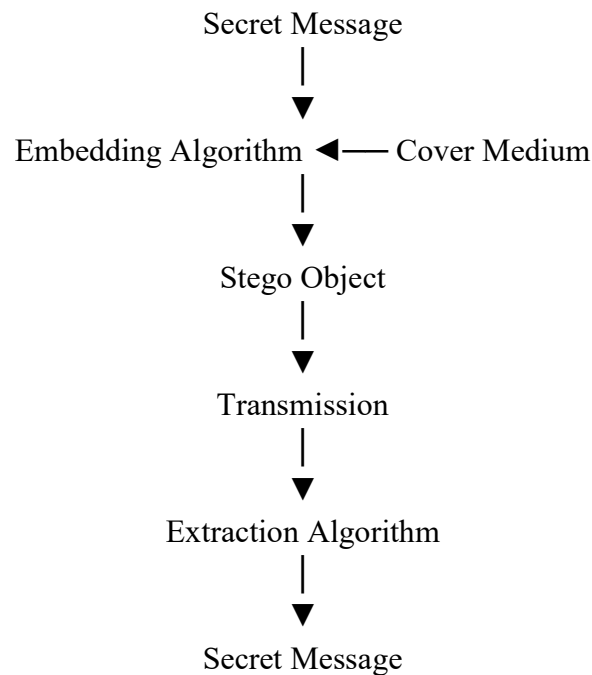
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3.5 Working Principle

Steganography involves two major stages:

1. **Embedding:** The sender hides the secret message inside a cover medium.
2. **Extraction:** The receiver retrieves the hidden message using the appropriate method or key.



Example

Alice wants to send:

`MEET AT 5 PM`

She embeds the message in a holiday photograph. The photograph appears normal to an ordinary viewer. Bob uses the appropriate extraction method to recover the hidden message.

3.6 Types of Steganography

A. Text Steganography

Information is hidden in ordinary text using techniques such as:

- Extra spaces
- Formatting patterns
- Capitalization
- Invisible characters
- Synonym substitution

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Advantages: Simple and suitable for short messages.

Limitations: Low capacity and potentially easy detection by text-analysis tools.

B. Image Steganography

Secret information is hidden by modifying image data. Common formats include BMP, PNG, JPEG, and TIFF.

LSB Technique

In the Least Significant Bit technique, small changes are made to the least significant bits of pixel values.

Example:

Original pixel : 11010010

Modified pixel: 11010011

Only one bit changes, so the visual difference may be imperceptible.

Applications: Secure communication, medical imaging, watermarking, and other information-hiding applications.

C. Audio Steganography

Information is embedded in audio signals using techniques such as:

- LSB
- Phase coding
- Echo hiding
- Spread spectrum

Applications include communication and copyright protection.

D. Video Steganography

A video provides many possible hiding locations, including:

- Individual frames
- Audio tracks
- Motion-related information
- Compression-related information

Its large capacity makes it suitable for multimedia applications.

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E. Network Steganography

Information is concealed within network communication, for example by manipulating:

- Packet timing
- Packet order
- Selected protocol fields
- Unused bits

It is relevant to cybersecurity research and covert-communication studies.

3.7 Classification Based on Key Usage

1. Pure Steganography

- No secret key is used.
- Security relies strongly on keeping the hiding method unknown.

2. Secret-Key Steganography

- Uses a shared secret key.
- Sender and receiver must know the key.

3. Public-Key Steganography

- Combines public-key cryptography with steganography.
- A public/private key pair can be used for secure information hiding and extraction.

3.8 Steganography vs Cryptography

Feature	Cryptography	Steganography
Objective	Hide the content of a message	Hide the existence of a message
Output	Ciphertext	Stego object
Visibility	Message appears encrypted	Message appears normal
Detection	Easy to notice encrypted data	Difficult to notice hidden data
Security	Based on encryption algorithm and key	Based on hiding technique and optionally a key
Example	AES, DES	LSB Image Hiding

Combined Use

In practice, cryptography and steganography are often used together:

1. Encrypt the message using AES.
2. Embed the encrypted message into an image.

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This provides two layers of protection:

- The message content is encrypted.
- The existence of the message is concealed.

3.9 Advantages

- Conceals the existence of communication
- Can be used with multimedia
- Can provide an additional layer of protection
- Useful for watermarking and ownership verification
- Can be combined with cryptography

3.10 Limitations

- Limited hiding capacity depending on the medium
- Compression or editing may destroy hidden information
- Steganalysis can identify statistical anomalies
- Some techniques require significant computation
- Discovery of the hiding method may expose the hidden information

3.11 Applications

- Cybersecurity research
- Covert communication
- Military and intelligence communication
- Digital watermarking
- Healthcare information embedding
- Multimedia security
- Digital forensics

3.12 AI and Steganography

AI can support both sides of information hiding:

AI-based steganography

- Learns ways to hide information with low visible distortion.
- Can support secure image sharing and other information-hiding applications.

AI-based steganalysis

- Attempts to detect hidden information by analyzing:
- Pixel distributions
- Statistical irregularities
- Textures
- Frequency-domain characteristics

Applications include digital forensics, cybersecurity, and security research.

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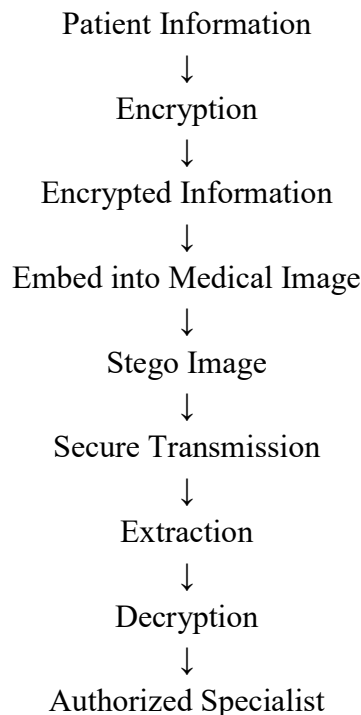
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3.13 Case Study: Medical Image Protection

A hospital needs to send an MRI scan and associated confidential information to a specialist.

A possible approach is:



This combines encryption with information hiding to protect sensitive information.

4. Perfect Secrecy and Its Limitations

4.1 Introduction

An important question in cryptography is whether an encryption system can provide security that does not depend on an attacker's computational limitations.

Claude Shannon introduced the mathematical concept of **perfect secrecy** in his work on secrecy systems. Perfect secrecy is also called **information-theoretic security**.

4.2 Definition

A cryptographic system provides **perfect secrecy** if observing the ciphertext provides no information about the plaintext.

This means that the probability distribution of the plaintext remains unchanged even after the ciphertext is observed.

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4.3 Example

Suppose the possible messages are:

- ATTACK
- RETREAT
- WAIT
- SURRENDER

If each has probability (0.25) before encryption and the same probabilities remain after observing a ciphertext, the ciphertext has provided no information about which message was sent.

By contrast, if observing the ciphertext changes the probabilities to something such as:

Message	Before	After
ATTACK	0.25	0.80
RETREAT	0.25	0.10
WAIT	0.25	0.05
SURRENDER	0.25	0.05

then the ciphertext provides information about the plaintext; therefore the system does not satisfy the perfect-secrecy condition.

4.4 Shannon's / Kerckhoffs' Principle

A secure cryptographic system should not depend on keeping the algorithm secret.

Principle: The security of a cryptographic system should depend on the secrecy of the key rather than the secrecy of the algorithm.

Therefore:

- The algorithm may be publicly known.
- The secret key must remain protected.

4.5 Conditions for Perfect Secrecy

The One-Time Pad achieves perfect secrecy when its required conditions are satisfied:

8. The key is truly random.
9. The key is at least as long as the message; for the classical OTP, it is equal in length.
10. The key is used only once.
11. The key remains secret.
12. The key is securely distributed.

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4.6 One-Time Pad (OTP)

The **One-Time Pad** is the standard example of a system proven to provide perfect secrecy when the above conditions are met.

Conceptually:

Plaintext + Random Key



Encryption



Ciphertext

The receiver uses the same one-time key to recover the plaintext.

The central requirement is **never to reuse the key**.

4.7 Mathematical Representation

Encryption:

$$C=E(K,P)$$

Decryption:

$$P=D(K,C)$$

Perfect secrecy:

$$P(P|C)=P(P)$$

- P = plaintext/message
- C = ciphertext
- P(P) = probability that a particular plaintext occurs **before** seeing the ciphertext
- P(P|C) = probability of that plaintext **after** seeing the ciphertext

4.8 Information-Theoretic vs Computational Security

Feature Perfect Secrecy Computational Security

Feature Perfect Secrecy Computational Security

Basis Mathematical / information-theoretic Computational difficulty

Unlimited computing power

Remains secure if conditions hold

May eventually be broken

Key

requirement

Very large key; OTP key is as long as the message

Generally much shorter and reusable

Example One-Time Pad (OTP) AES and other practical algorithms

Practicality Difficult for large-scale use Highly practical

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4.9 Limitations of Perfect Secrecy

1. Key Length

The OTP requires a random key comparable in length to the message.

2. Random Key Generation

Truly random keys are difficult to generate and maintain at scale.

3. Secure Key Distribution

The key must reach the receiver securely before use.

4. Storage

Large communication volumes require correspondingly large amounts of key material.

5. Key Management

Every message requires fresh key material, making management difficult for large organizations.

6. No Key Reuse

Reusing an OTP key can reveal relationships between ciphertexts and destroy the perfect-secrecy property.

7. Synchronization

Sender and receiver must correctly coordinate the key material used for each message.

8. Operational Cost

Generating, distributing, storing, protecting, and destroying large quantities of key material is expensive and complex.

4.10 Practical Example

If a bank must securely process 10 TB of information per day using an OTP, it would need approximately 10 TB of fresh random key material per day, in addition to securely distributing and managing that key material.

This illustrates why practical systems generally use computationally secure encryption such as AES instead.

4.11 Applications

Perfect-secrecy techniques can be appropriate in specialized situations where the amount of data is limited and the value of the information justifies extensive key management, such as certain military, intelligence, or highly sensitive command communications.

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4.12 Perfect Secrecy and AI

AI systems often process enormous datasets, making OTP-style key management impractical for general-purpose use. Practical AI systems instead rely on computationally secure mechanisms, secure communication protocols, and appropriate key-management systems.

5. Shannon's Principles: Foundation of Modern Cryptography

5.1 Introduction

Claude E. Shannon's 1949 work, **Communication Theory of Secrecy Systems**, established a mathematical foundation for analyzing secure communication.

Important concepts associated with Shannon's work include:

- Perfect secrecy
- Information-theoretic security
- Confusion
- Diffusion
- Entropy

These ideas influenced the design of modern block ciphers.

5.2 Claude Shannon

Claude Shannon was an American mathematician, electrical engineer, and computer scientist, widely known as the **Father of Information Theory**.

His contributions relevant to cryptography include:

- Mathematical analysis of secrecy
- Perfect secrecy
- Confusion and diffusion
- Entropy and information uncertainty
- Foundations for secure communication

5.3 Confusion

Definition: Confusion makes the relationship between the **secret key** and the **ciphertext** as complex as possible.

The aim is to make it difficult for an attacker to infer the key from the ciphertext.

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Example

For the same plaintext:

HELLO + Key 1 $\rightarrow$ Ciphertext A

HELLO + Key 2 $\rightarrow$ Ciphertext B

Changing the key should produce a substantially different ciphertext.

Characteristics

- Makes key relationships difficult to identify.
- Reduces predictable patterns.
- Commonly associated with substitution operations.
- In AES, **SubBytes** provides the principal nonlinear substitution that contributes to confusion.

5.4 Diffusion

Definition: Diffusion spreads the influence of plaintext information across many ciphertext bits.

The purpose is to hide statistical relationships between plaintext and ciphertext.

Example

A small change in plaintext should result in changes throughout the ciphertext:

Plaintext 1 $\rightarrow$ Ciphertext 1

Plaintext 2 $\rightarrow$ Very different Ciphertext 2

Characteristics

- Hides plaintext patterns.
- Reduces statistical information.
- Makes frequency-based analysis more difficult.
- Achieved through permutation and mixing operations.
- In AES, ShiftRows and MixColumns provide important diffusion effects.

5.5 Confusion vs Diffusion

Confusion	Diffusion
Hides relationship between key and ciphertext	Hides relationship between plaintext and ciphertext
Achieved through substitution	Achieved through permutation and mixing
Makes key prediction difficult	Makes pattern detection difficult
Increases key complexity	Spreads information across the ciphertext

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5.6 Entropy

Entropy measures uncertainty, randomness, or unpredictability.

In cryptographic contexts:

- **High entropy** → less predictable information
- **Low entropy** → more predictable information

Example:

`123456` has low unpredictability, while a long, randomly generated secret value has much higher unpredictability.

Entropy is important for:

- Random key generation
- Session keys
- Password protection
- Cryptographic protocols
- Digital signatures

5.7 Avalanche Effect

The **Avalanche Effect** is the desirable property that a small change in plaintext or key causes a large, unpredictable change in ciphertext.

Example:

Plaintext 1: HELLO

Plaintext 2: JELLO

Even though only one character changes, a well-designed cipher should produce substantially different ciphertexts.

The avalanche effect results from the combined effect of substitution, mixing, and repeated cipher rounds.

5.8 Product Cipher

A **product cipher** combines multiple simple cryptographic transformations to produce a stronger cipher.

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The basic idea is:

Substitution → Permutation/Mixing → Repeated Rounds

Modern block ciphers such as AES use repeated transformations that embody this general design philosophy.

5.9 Shannon's Principles in AES

AES Transformation Shannon Principle

SubBytes Confusion

ShiftRows Diffusion

MixColumns Diffusion

AddRoundKey Key mixing

AES therefore combines substitution, mixing, permutation-like operations, and key addition over multiple rounds.

5.10 Relevance to AI

AI systems process sensitive information such as:

- Medical records
- Financial data
- Biometric information
- Autonomous-vehicle data
- Training datasets
- Cloud-based AI workloads

Secure cryptographic algorithms based on sound design principles help protect these systems.

Applications include:

- Secure AI model deployment
- Privacy-preserving machine learning
- Federated learning
- Secure cloud AI
- Protection of AI training data

5.11 Case Study: Cloud-Based AI Healthcare

A hospital uploads MRI images to a cloud platform for AI-based diagnosis.

Security risks:

- Unauthorized access

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- Interception
- Disclosure of patient information

Protection:

- Encrypt the data using a strong symmetric algorithm such as AES.
- Use secure key management and access control.
- Protect communication channels with appropriate security protocols.

AES uses nonlinear substitution and repeated mixing transformations to make cryptanalysis more difficult.

5.12 Advantages of Shannon's Design Principles

- Increase resistance to statistical analysis
- Reduce exploitable relationships between inputs, keys, and outputs
- Provide a foundation for modern block-cipher design
- Support secure processing of large datasets
- Apply to systems used in banking, healthcare, cloud computing, IoT, and AI

5.13 Limitations

Shannon's principles are **design principles**, not a complete encryption algorithm. Secure implementation still requires:

- Strong algorithms
- Appropriate key lengths
- Secure key management
- High-quality randomness
- Protection against implementation and side-channel attacks

6. Security Attacks in Cryptography

6.1 What is a Security Attack?

A **security attack** is an attempt to gain unauthorized access to information, modify data, disrupt communication, impersonate a legitimate user, or compromise a cryptographic system.

An attacker may attempt to:

- Read information
- Modify information
- Impersonate a user
- Replay a valid message

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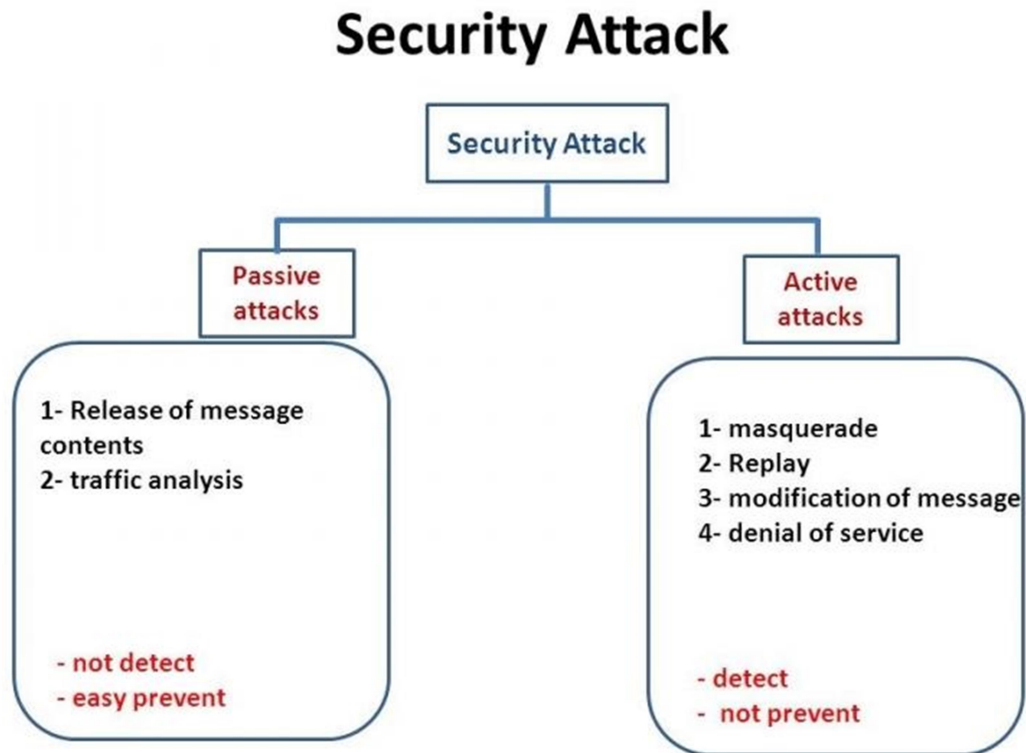
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- Prevent service availability
- Discover secret information

6.2 Classification

Security attacks are broadly classified into:



6.3 Passive Attacks

In a **passive attack**, the attacker observes or listens to communication but does not alter the transmitted information.

A. Release of Message Contents

The attacker obtains the actual contents of a communication.

Example:

Alice sends:

Alice $\longrightarrow$ Bob
"PAY ₹10,000"

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Eve secretly listens:



If Eve can obtain the contents, confidentiality has been compromised.

Primary protection: Encryption and secure communication.

B. Traffic Analysis

Even if message contents are encrypted, an attacker may observe:

- Who is communicating
- Frequency of communication
- Message size
- Time of communication
- Amount of traffic

The attacker may infer useful information from these patterns without reading the actual content.

Key idea:

Passive attacks - observe but do not modify.

6.4 Active Attacks

In an **active attack**, the attacker modifies, inserts, deletes, replays, or disrupts communication.

Four major types are discussed below.

A. Masquerade Attack

A masquerade occurs when an attacker **pretends to be a legitimate user**.

Example:

Actual:

Eve —————▶ Bob

"I am Alice"

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A hacker using stolen credentials to access another person's account is an example of masquerade.

Masquerade = Impersonation

B. Replay Attack

A replay attack occurs when an attacker captures a valid message and transmits it again later.

Example:

Original:

Alice —————▶ Bank

"Transfer ₹10,000 to Bob"

Captured by Eve

↓

Replayed later

↓

Eve —————▶ Bank

"Transfer ₹10,000 to Bob"

Protection mechanisms include:

- Nonces
- Timestamps
- Sequence numbers

C. Modification of Messages

The attacker changes a legitimate message before it reaches the receiver.

Example:

Original:

"Transfer ₹1,000 to Ravi"

Modified:

"Transfer ₹10,000 to Ravi"

This directly threatens integrity.

Protection can include:

- Message Authentication Codes (MACs)
- Digital signatures
- Authenticated encryption

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D. Denial of Service (DoS)

A DoS attack attempts to prevent legitimate users from accessing a service.

Example:

Attacker —► Server

Attacker —► Server

Attacker —► Server

Attacker —► Server

↓

Server overloaded

↓

Legitimate user cannot access service

A flood of requests can consume server resources and reduce availability.

6.5 Summary of Security Attacks

Attack	What attacker does	Example
Release of message contents	Reads information	Reads an encrypted communication after breaking encryption
Traffic analysis	Observes communication patterns	Determines who is communicating and when
Masquerade	Pretends to be someone else	Hacker logs in using stolen credentials
Replay	Captures and resends a valid message	Repeats a payment request
Modification	Changes a message	₹1,000 → ₹10,000
Denial of Service	Prevents access to service	Floods a server with requests

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7. Unit-Level Review Questions

Short Answer Questions

1. Define cryptography.
2. What is plaintext?
3. What is ciphertext?
4. Explain encryption and decryption.
5. State the security objectives of cryptography.
6. Define a symmetric cipher.
7. Why is symmetric encryption called secret-key cryptography?
8. List the components of the symmetric cipher model.
9. Write the mathematical representation of symmetric encryption and decryption.
10. What is the key distribution problem?
11. Name common symmetric encryption algorithms.
12. What is steganography?
13. Differentiate a cover medium and a stego object.
14. Explain the LSB technique.
15. What is network steganography?
16. Differentiate cryptography and steganography.
17. Define perfect secrecy.
18. State the conditions required for the One-Time Pad.
19. What is information-theoretic security?
20. Why must an OTP key not be reused?
21. What are the limitations of perfect secrecy?
22. Who was Claude Shannon?
23. Define confusion.
24. Define diffusion.
25. What is entropy in cryptography?
26. Explain the Avalanche Effect.
27. What is a product cipher?
28. How do SubBytes, ShiftRows, and MixColumns contribute to AES?
29. What is a security attack?
30. Differentiate passive and active attacks.
31. What is a masquerade attack?
32. What is a replay attack?
33. What is modification of messages?
34. What is a Denial-of-Service attack?
35. What is traffic analysis?

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Long Answer Questions

13. Explain cryptography, its objectives, applications, and challenges.
14. Describe the Symmetric Cipher Model with a suitable diagram and example.
15. Discuss the advantages, disadvantages, and applications of symmetric encryption.
16. Compare symmetric and asymmetric cryptography.
17. Explain steganography, its working principle, types, and applications.
18. Explain image steganography and the LSB technique.
19. Compare cryptography and steganography with suitable examples.
20. Explain perfect secrecy and the One-Time Pad.
21. Discuss the practical limitations of perfect secrecy.
22. Compare perfect secrecy with computational security.
23. Explain Claude Shannon's contribution to modern cryptography.
24. Discuss confusion and diffusion with examples.
25. Explain entropy and the Avalanche Effect.
26. Explain the product-cipher concept and its relation to modern block ciphers.
27. Explain how AES reflects Shannon's design principles.
28. Explain passive and active security attacks with examples.
29. Discuss release of message contents and traffic analysis.
30. Explain masquerade, replay, modification, and DoS attacks.
31. Discuss the role of cryptography in securing Artificial Intelligence applications.

Previous-Year / University Questions(Unit-1)

1. What is Cryptology?
2. Transform the message “**happy birthday to you**” using the Rail Fence technique.
3. Transform “**Cryptography and Network Security**” using simple columnar transposition with the order **4, 6, 1, 2, 5, 3**.
4. (a) Encrypt “**I Love Cryptography**” using the Caesar cipher.
(b) Draw the model for network security and explain it.
5. Encrypt a given message using the Playfair cipher with a standard Playfair key.
6. Define cryptography, cryptology, and cryptanalysis.
7. Differentiate symmetric-key and asymmetric-key cryptography.
8. Write short notes on any two: Playfair cipher with example, transposition techniques, and steganography.
9. Explain the basic models of symmetric cipher cryptography techniques.
10. Explain transposition techniques with suitable examples.
11. Explain security attacks and their types.
